OBSTACLE DETECTION CAR USING ARDUINO AND ULTRASONIC SENSOR

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Submitted by

1. BHUKYA VEDA VANGMAI (23B81A0410)

2. MORE VAISHNAVI (23B81A0457)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CVR COLLEGE OF ENGINEERING

(An Autonomous Institution with Grade 'A' & Affiliated to JNTUH)
Mangalpalli (V), Ibrahimpatnam (M), Ranga Reddy (D), Telangana – 501510

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ABSTRACT

This project involved designing and building an obstacle-detecting car using an Arduino Uno, L298N motor driver, ultrasonic sensor, servo motor and four DC motors. The main aim is to create a small vehicle that could move on its own while avoiding objects in its path. The ultrasonic sensor mounted on a servo motor played a key role by scanning the front area in three directions— left, right, and center. This scanning method helped the system decide the direction with the most space available for movement.

The Arduino acted as the central controller taking readings from the sensor and then controlling the motors through the motor driver. Power for the DC motors was supplied through an external battery while the Arduino and other components were powered by a 5V source. The program was designed so that the car would keep moving forward unless it detected an obstacle closer than a certain distance.

When an obstacle was detected the servo motor would rotate the ultrasonic sensor to measure distances on each side. Based on these readings, the Arduino decided which direction had fewer obstacles and then turned the car accordingly. This process allowed the car to move around while avoiding objects in real time and continue moving safely in open directions.

The final result was successful in showing that the car could move on its own and avoid obstacles. It could stop when needed, scan its surroundings, and choose a clear path to continue. The project met its goals and demonstrated how sensors, motors, and simple decision-making code can be used together for automatic movement. It also provided hands-on experience with hardware and programming, and sets a good base for future upgrades.

1. INTRODUCTION

With the rapid advancement of technology, automation has become an essential part of modern engineering systems. One of the most significant developments in this field is the emergence of autonomous vehicles in robotics. This project titled **“Obstacle-Avoiding Car”** focuses on designing and developing a small-scale robotic vehicle capable of detecting and avoiding obstacles without human intervention.

The system is built using an **Arduino Uno microcontroller, ultrasonic sensor, servo motor, L298N motor driver, and DC motors**. It simulates the working principle of real-time autonomous navigation systems by detecting surrounding objects and making intelligent movement decisions. This project serves as a practical application of embedded systems, robotics, and sensor-based automation.

2. AIM OF THE PROJECT

The main aim of this project is to design and develop an autonomous robotic car that can move independently while avoiding obstacles in its path using real-time sensor data. The system is designed to detect objects, stop when an obstacle is found, scan the surroundings (left, right, and center), and select the safest direction for movement. This demonstrates the integration of sensors, actuators, and control logic for basic autonomous navigation.

3. METHODOLOGY

The development of the obstacle-avoiding car involved selecting and integrating key hardware components such as the **Arduino Uno, HC-SR04 ultrasonic sensor, servo motor, L298N motor driver, and DC motors**. The ultrasonic sensor is mounted on a servo motor to enable directional scanning.

During operation, the car moves forward continuously until an obstacle is detected within a predefined distance. Once detected, the system stops and the servo rotates the ultrasonic sensor to scan the left, right, and center directions. Based on the measured distances, the Arduino processes the data and determines the direction with the most free space. The motor driver then controls the DC motors to turn the vehicle accordingly.

The system is powered using an external battery pack for the motors, while the Arduino and control circuitry are powered through a regulated 5V supply. The program is implemented using the Arduino IDE, enabling real-time decision-making and autonomous navigation.

4.HARDWARE SPECIFICATIONS

The obstacle-avoiding car is built using the following hardware components:

- **Arduino Uno** – Acts as the main microcontroller that processes sensor data and controls the system.
- **Ultrasonic Sensor (HC-SR04)** – Used for detecting obstacles by measuring distance using sound waves.
- **Servo Motor** – Rotates the ultrasonic sensor to enable directional scanning (left, right, and center).
- **L298N Motor Driver Module** – Controls the direction and speed of the DC motors.
- **4 × DC Motors, Wheels, and Chassis** – Provide movement and mechanical structure for the robotic car.
- **Battery (9V or 12V depending on motor requirements)** – Supplies power to the motors and control circuit.

SOFTWARE SPECIFICATIONS

- **Arduino IDE** – Used for writing, compiling, and uploading the program code to the Arduino Uno.

5. SIGNIFICANCE OF THE WORK

The significance of this project lies in demonstrating the fundamental principles of **autonomous robotic systems using embedded technology**. It provides a practical model of how machines can sense their environment, process information, and make decisions without human intervention.

This project is highly relevant to real-world applications such as **self-driving vehicles, industrial automation systems, warehouse robots, and surveillance robots**, where obstacle detection and avoidance are essential for safe operation.

In addition, the system offers hands-on experience in integrating **sensors, microcontrollers, motor drivers, and control algorithms**, making it particularly valuable for students and beginners in robotics, automation, and embedded systems. It strengthens understanding of real-time decision-making and hardware–software interaction, which are core concepts in modern engineering applications.